

The Jingzhang Diagram

Centennial Jingzhang AI Innovation Belt Urban Design Proposal (Final Evolution v1.4)

A city that runs by the diagram - Spatial Operating System & Four-System Hardwired Governance

Author: Juwan-Hwang • Formal AI urban design package

SEB v0.5.0 tabletop self-check complete (participant-declared, no third-party certification)

Provisional process boundary (EPSG:4548); not an official planning approval redline

THE JINGZHANG DIAGRAM: OVERALL SPATIAL OPERATING SYSTEM

PROVISIONAL BOUNDARY (NON-STATUTORY)

Four Epochs of Innovation along the Corridor & Three Key Areas of the Spatial OS

I. FOUR INNOVATION EPOCHS

1909 Jingzhang Railway

First trunk railway independently surveyed, designed and built by China; switchback topology beats the grade
[National Railway Museum Records]

1980s Zhongguancun

Historical narrative milestone of Chinese tech entrepreneurship and institutional innovation
[Historical Narrative Milestone]

2019 Jingzhang High-Speed Railway

World's first 350 km/h intelligent high-speed railway (full qualifier); intelligent running & Beidou at 350 km/h
[Ministry of Transport Official Statement]

2026 Centennial AI Innovation Belt

Proposal design proposition: human-AI co-ascent; diagram governance & The Human Switch safety
[Design Concept Proposition]

II. SPATIAL OPERATING SYSTEM - THREE KEY AREAS

Zhongzhiyuan AI Testbed

192.1 ha • Microgrid compute
Autonomous loop test track

Provisional boundary polygon • EPSG:4548 model internal only

Beijing AI Origin Community

104.3 ha • Staffed dispatch hall
Living district & night school

Provisional boundary polygon • EPSG:4548 model internal only

Dazhongsi Marshalling Yard

72.0 ha • Commercialisation
Two-part handover desk

Provisional boundary polygon • EPSG:4548 model internal only

Four industry test & validation scenarios:

Autonomous safety corridor

Robot city-service yard

LLM governance sandbox

Heritage digital twin

III. DIAGRAM GOVERNANCE: 4 HARDWIRED INTERLOCK SYSTEMS

信号制 Signal

Multi-tiered dynamic permissions
Charter 10 human-first principles

闭塞制 Block

Spatial & compute separation
Charter 10 human-first principles

联锁制 Interlock

SEB v0.5.0 tabletop self-check 7/7 (participant-declared)
Charter 10 human-first principles

路票制 Token

Human Switch tiered interface & opt-out
Charter 10 human-first principles

submissions/Juwan-Hwang/jingzhang-diagram • geometry/*.geojson • metrics.json • 图表为过程性研究表达，非法定依据

LAND USE STRUCTURE & ZONING PARTITION (TOPOLOGY SEAMLESS)

11.41 km² Provisional Site Area · Green Ratio 15.68% · Public Space Ratio 9.31%

PROVISIONAL GEOMETRY · EPSG:4548

LAND USE TOPOLOGY PARTITION

0802 Research & AI Innovation (Zhongzhiyuan)

192.1 ha (AI Testbed Loop)

05 Commercial & Commercialisation (Dazhongsi)

72.0 ha (Marshalling Yard)

1403 Public Plazas & Civic Rooms

106.23 ha (Public Space 9.31%)

0701 Residential & Living (AI Origin Community)

104.3 ha (Living District)

1401 Green Space & Parks (Unary Union)

178.94 ha (Green Ratio 15.68%)

1207 Transport & Slow Mobility Corridors

12.8 km Continuous Network

METRIC RECALCULATION

Total Site Area

11,412,825.39 m² (11.41 km²)

Green Space Union Area

1,789,421.32 m² (178.94 ha)

Green Ratio

0.15679 (15.68%)

Public Space Union Area

1,062,301.12 m² (106.23 ha)

Public Space Ratio

0.09308 (9.31%)

Key Areas Total Area

368.40 ha (3 Key Areas)

Floor Area Ratio (FAR)

unknown (Pending Statutory Masterplan)

Topology Partition Integrity

PASS (0 gaps / 0 overlaps)

THREE KEY DETAILED-DESIGN AREAS (PROVISIONAL BOUNDARIES)

Zhongzhiyuan (192.1 ha) · AI Origin Community (104.3 ha) · Dazhongsi (72.0 ha)

PROVISIONAL GEOMETRY · EPSG:4548

Zhongzhiyuan AI Testbed (192.1 ha)

ZHONGZHIYUAN AI ACCELERATION AREA

- Positioning: Autonomous mobility & low-altitude testbed
- Facilities: Microgrid compute hub, closed test track
- Governance: L2 closed paired testing, power monitoring
- Boundary: Provisional polygon awaiting statutory redline

TECHNICAL DISCLOSURE

- Coordinate: EPSG:4548
- Area Tolerance: ±0.0% aligned
- Scope: Conceptual design
- Condition: Statutory permits

Beijing AI Origin Community (104.3 ha)

BEIJING AI ORIGIN COMMUNITY

- Positioning: Origin dispatch & inclusive living district
- Facilities: Physical dispatch window, night school, greenway
- Governance: L0/L1 baseline, 100% staffed window floor
- Boundary: Qinghuayuan heritage setback CONS-001

TECHNICAL DISCLOSURE

- Coordinate: EPSG:4548
- Area Tolerance: ±0.0% aligned
- Scope: Conceptual design
- Condition: Statutory permits

Dazhongsi AI Industry Cluster (72.0 ha)

DAZHONGSI AI INDUSTRY CLUSTER

- Positioning: Marshalling yard & commercialisation hub
- Facilities: Heritage civic plaza, 4-station service desk
- Governance: SEB v0.5.0 four-station sequence enforced
- Boundary: Dazhongsi heritage control zone CONS-002

TECHNICAL DISCLOSURE

- Coordinate: EPSG:4548
- Area Tolerance: ±0.0% aligned
- Scope: Conceptual design
- Condition: Statutory permits

SLOW MOBILITY, TRANSIT & BLUE-GREEN PUBLIC SPACE NETWORK

PROVISIONAL GEOMETRY • EPSG:4548

12.8 km Continuous Greenway • Pedestrian Priority • Barrier-Free Access • The Human Switch (Tiered Safety Interface)

1. MULTI-TIERED MOBILITY NETWORK

Spine Continuous Greenway

12.8 km linear park connecting Beijing North to Tsinghua

AI Micro-Shuttle Test Lane

Speed cap < 15 km/h, dedicated lane, pedestrian priority

Barrier-Free & Silver Walkways

100% tactile paving, gentle slopes, emergency call points

Ring-Road Stitching Bridges

Bridging Ring 2/3/4 barriers for uninterrupted mobility

2. BLUE-GREEN SYSTEM & THE HUMAN SWITCH

Xiaoyuehe Sponge Green Infrastructure

178.94 ha green space, stormwater retention, microclimate

Origin Civic Plazas & Open Space

106.23 ha public space for community life and events

The Human Switch — Tiered Safety Interface

Public alert logged & displayed -> authorised on-site e-stop (3 s hold + two-person confirm) -> professional recovery at dispatch hall; fully audited with accountability chain (conceptual design, no ready-made safety claim)

SEB v0.5.0 Hardwired Self-Check

Two-part handover ledger, staffed windows; 7/7 tabletop fixtures match (participant-declared, synthetic fixtures, no third-party certification)

QUANTITATIVE METRICS, SIMULATION LEDGER & EVIDENCE CROSSWALK

PROVISIONAL GEOMETRY • EPSG:4548

Spatial Recalculation • 10 Synthetic Tasks (90% Success) • SEB v0.5.0 Tabletop Self-Check (7/7 Match) • 7-Dimension Rubric

1. SPATIAL & SIMULATION METRICS

Metric Key	Declared Value	Provenance / Formula
Total Site Area (site_area_sqm)	11,412,825.39 m ²	EPSG:4548 provisional boundary
Green Space Area (green_space_area_sqm)	1,789,421.32 m ²	unary_union intersection
Green Ratio (green_ratio)	0.15679 (15.68%)	Exact ratio (topology-consistent)
Public Space Area (public_space_area_sqm)	1,062,301.12 m ²	unary_union intersection
Public Space Ratio (public_space_ratio)	0.09308 (9.31%)	Exact ratio (topology-consistent)
Simulation Task Count	10 tasks (SIM-001-010)	Deterministic seed: 20260831
Simulation Success Rate	0.90 (90.0%)	Honest disclosure of overrun
SEB Tabletop Self-Check	7 / 7 fixtures match (Exit 0)	Participant-declared • synthetic fixtures • no third-party certification

指标与 simulation.json / metrics.json 机器可读值一致；SEB 推演为参与者桌面自检，可离线复跑，未经第三方认证

2. RUBRIC CROSSWALK & DATA GAP DISCLOSURE

Brief Alignment (20%)

3-level scopes • 6 required tasks • 3 key areas • 12.8 km corridor

Originality (10%)

Working Diagram concept • Switchback unit • 4 innovation epochs

AI Innovation (15%)

4-system interlock • 12 scenario cards • SEB v0.5.0 self-check 7/7

Feasibility (20%)

JZ-01~10 lifecycle ledger • Adopter rights • 4-phase rollout

Public Interest (10%)

100% staffed window floor • Barrier-free • The Human Switch (tiered)

Risk Compliance (10%)

Provisional boundaries • Heritage setbacks • Adverse readings

Completeness (15%)

Bilingual parity • Embedded fonts • A3/A0 drawings • 4 Gate PASS

Statutory Notice: Official redlines & FAR are unknown; conceptual design does not replace statutory permits.